

100 Days of Linux - Day 079

Title: Network Troubleshooting Workflow

Today's Goal

Learn a simple step-by-step process for troubleshooting common Linux network problems using the commands learned so far.

Why This Matters

In real jobs, engineers rarely use a single command. They combine several tools to identify and fix connectivity issues quickly.

Workflow of the Day

1. Check connectivity with **ping**.
2. View IP information with **ip addr**.
3. Check routes using **ip route**.
4. Test a website using **curl -I**.
5. Download a file using **wget** if required.

Beginner Lesson

Always start with the simplest check. If ping fails, inspect your IP address and routing table. If the network is working but a website is not responding correctly, inspect its headers with curl. Use wget only when you need to download files.

Practice Questions

1. Verify internet connectivity with 4 ping requests.
2. Display your IP addresses.
3. Display your routing table.
4. Show the HTTP headers of <https://example.com>.
5. Download a sample page using wget and verify it with ls.

Hints

Think about which command answers each troubleshooting question before typing it.

Mini Challenge

Create a file named `troubleshooting_steps.md` and write the five commands in the correct troubleshooting order.

Common Mistakes

Skipping basic connectivity checks, confusing curl with wget, or ignoring routing information.

Did You Know?

Experienced Linux administrators often solve network issues by following the same repeatable workflow every time.

Pro Tip

Use a consistent troubleshooting checklist instead of guessing.

Answer Key (Instructor Only)

Q1: `ping -c 4 google.com`

Q2: `ip addr`

Q3: `ip route`

Q4: `curl -I https://example.com`

Q5: `wget https://example.com` then `ls`

Homework

Repeat today's troubleshooting workflow without looking at your notes.

Next Day Preview

Tomorrow you'll complete the networking module with a mini project.